



COMSATS University Islamabad WAH Campus

Object Oriented Programming Department of Computer

Members: Sami Shah FA25-BCS-109

Huzaifa Ahmed FA25-BCS123

Khubaib Ismail FA25-BCS-165

Class: BCS-1C

Professor: Miss Anum Ilyas

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Assignment 2 - Service and Deployment Models of Cloud Computer

Introduction:

The gig economy is growing rapidly, but local and university level freelancers lack a dedicated and accessible platform. International platforms often charge high commission fees and present complex payment gateway challenges for local users. This project provides a localized digital marketplace to seamlessly connect skilled students and professionals with clients needing short term project assistance.

Problem Statement:

Local freelancers and clients struggle to find a streamlined, secure, and cost effective environment to negotiate digital services. Existing major platforms do not adequately cater to micro jobs or university level skill exchanges. There is a missing system to actively manage the business lifecycle locally from proposal submission to secure escrow payments and milestone tracking.

Proposed Solution:

We will develop a comprehensive Management Information System in Java that functions as an active digital marketplace. Users will interact with the system through a rich graphical user interface built entirely with Java Swing. The application will implement distinct dashboards tailored to each user role, ensuring intuitive navigation for job posting, bidding, and wallet management.

OOP Principles:

- We will secure all user data, wallet balances, and contract details using private attributes accessed only through designated getters and setters.
- We will create a base User class that extends into specific Freelancer, Client, and Admin derived classes.
- We will utilize method overriding for flexible behaviors, such as rendering completely different dashboard interfaces based on the user type logged in.
- We will implement abstract classes or interfaces to define common behaviors like transaction processing or dispute resolution without exposing the complex underlying logic to the main application.

System Design:

- The system will include key classes like User, Freelancer, Client, Admin, JobListing, Proposal, Contract, and Wallet.
- The Freelancer and Client classes will interact primarily through Proposal and Contract objects to update statuses based on milestone submissions and approvals.
- GUI classes will strictly handle the Java Swing visual components to keep the visual interface separated from the core business logic.
- A comprehensive class diagram will be provided to map out all inheritance paths and class interactions.

Functional Requirements:

- Freelancers will be able to create dynamic portfolios, browse job boards, submit customized bids, manage incoming contract offers, submit milestone deliverables, and withdraw from virtual wallets.
- Clients will be able to post new job listings, fund escrow accounts, review and accept incoming bids, and approve submitted milestones.
- Administrators will be able to resolve dispute tickets raised by users, suspend fraudulent accounts, adjust platform commission rates, and view global transaction analytics.